DERICK BESSA

FULL-STACK DEVELOPER

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PROFESSIONAL SUMMARY

Full-Stack Developer with experience in React, Next.js, Angular, and React Native on the frontend; Node.js, Java/Spring Boot, and Python/FastAPI on the backend. Applied research background in computer vision, integrating AI models into end-to-end mobile and web systems. Founder of **TáNaLista**, an event management SaaS with QR Code check-in.

EDUCATION & LANGUAGES

B.Sc. in Computer Science – IFCE (Federal Institute of Ceará), Fortaleza 2024 - 2028

Languages: English (Advanced) · Portuguese (Native)

TECHNICAL SKILLS

`Frontend:` React.js, Next.js, Angular, React Native, TypeScript, Tailwind CSS

`Backend:` Node.js, Java/Spring Boot, Python/FastAPI, REST APIs, JWT/OAuth2

`Databases:` PostgreSQL, SQLite, MySQL

`AI & Computer Vision:` YOLOv8/v11, OpenCV, MediaPipe, PaddleOCR, scikit-learn, pandas

`DevOps & Tools:` Docker, Git, GitHub Actions, Vercel, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

LabVicia – IFCE, Fortaleza Sep 2022 - Present

Full Stack Developer · Research Scholar

- Implemented server-side pagination and eliminated redundant API calls, reducing data transfer volume and making table loading noticeably faster for end users.
- Refactored the inbox module with infinite scroll and automatic message updates, eliminating manual refreshes and making the communication flow continuous and uninterrupted.
- Delivered full internationalization for 4 languages (PT, EN, ES, DE) with dynamic switching via ngx-translate, enabling international teams to use the system with no rebuild or redeployment required.
- Standardized components with Angular Material across the entire system, reducing visual inconsistencies and speeding up new screen development for the team.

LAPISCO – Research Lab at IFCE, Fortaleza Jun 2023 - Mar 2026

Research Scholar · Computer Vision & Full Stack Development

- Integrated detection and classification models (YOLOv8/v11, MediaPipe, OpenCV) with the Java/Spring Boot backend via a Python/FastAPI layer, enabling real-time processing of frames captured by the camera in the React Native app.
- Built dashboards and technical reports using pandas and scikit-learn to monitor model training, allowing the team to identify bottlenecks and tune hyperparameters based on concrete data.
- Refactored Java backend APIs with a focus on separation of concerns and object-oriented design, improving maintainability and reducing onboarding time for new team members.
- Worked across the full AI pipeline — from image capture in the React Native frontend to result consolidation in the Java backend — ensuring traceability and data consistency across all layers.

Gomes & Araújo Law Firm – Freelance 2025

Front-End Developer

- Delivered a complete landing page in React 19 + Vite + Tailwind CSS with 8 sections and a fully responsive layout, with no external UI library dependencies, resulting in a lighter bundle and faster load times.
- Built a Node.js/Express backend with transactional email delivery (Resend API) and a WhatsApp CTA, converting page visitors into leads directly without any external CRM tool.

PROJECTS

SFA – Automotive Inspection System (License Plate Reader)

Automatic vehicle license plate recognition

- Achieved 97% accuracy in license plate recognition by combining detection via YOLOv8 (custom-trained model) with OpenCV preprocessing and PaddleOCR reading, making the system reliable for real-world enforcement use.
- Implemented support for three plate formats (old standard, Mercosul, and motorcycle) with per-format character correction logic and automatic image rotation, eliminating failures caused by incorrect camera angles.
- Exposed the pipeline as a Python/FastAPI layer integrated with the Java/Spring Boot backend, allowing computer vision processing to run in a decoupled manner with no impact on the main system's performance.

Libras Translator

Brazilian Sign Language recognition from static images

- Built a complete ML pipeline in Python — collecting a custom dataset with 26 classes, extracting hand landmarks via MediaPipe Hands, and training a Random Forest classifier with scikit-learn — delivering a functional model for recognizing the Libras alphabet.
- Applied relative coordinate normalization to ensure scale and position invariance, making the model robust to variations in camera distance and framing without the need for retraining.
- Implemented an inference module with OpenCV to visualize landmarks and overlay predictions on the image, enabling fast visual model validation without a dedicated interface.

TáNaLista

Event management SaaS with QR Code check-in

- Built a SaaS platform in Node.js, Express, and PostgreSQL with Prisma ORM for QR Code check-in and real-time guest tracking, eliminating manual lists and reducing event entry time for organizers.
- Implemented JWT authentication with bcrypt and a modular architecture, ensuring secure access control and enabling each system module to evolve independently.
- Configured CI/CD with GitHub Actions and automated deployment to Vercel, eliminating manual deploys and ensuring every merge to main is immediately live in production.